

Rectified Flow for Chromosome Detection: Stable Coupling and Few-Step Inference

Anonymous submission

Abstract

Diffusion-based object detectors achieve competitive performance but suffer from slow inference due to curved denoising trajectories. *Rectified Flow* (RF) replaces the stochastic Denoising Diffusion Probabilistic Models (DDPM) process with straight-line ODE paths, enabling efficient few-step inference. We present the first systematic study of RF for chromosome karyotyping, making three contributions. **(1) RF training enables effective few-step detection:** on the 24-object benchmark, RF achieves mean Average Precision (mAP) 0.856, a +0.082 mAP improvement over the DDPM Euler baseline (0.774), and our best variant (DPM-Solver++, mAP 0.863) achieves +0.076 mAP over DiffusionDet. A solver $\times$ step disentanglement ablation reveals that 94% of this gain stems from the RF training paradigm (straight-line ODE paths + shifted schedule), with solver type and step count contributing only 6%. **(2) Stochastic Coupling (StochOT) stabilizes training:** we theoretically characterize Optimal Transport (OT) Diversity Collapse in low-dimensional detection space ($\Delta H \leq \log K$, severity $\Delta H/H \approx 0.69$) and propose a stochastic coupling via Sinkhorn transport. While the mAP gain is marginal (+0.002), Stochastic Coupling reduces within-run late-training epoch-level mAP oscillation by $4.6\times$ (epoch std 0.006 $\rightarrow$ 0.0013). **(3) DPM-Solver++ enables 2-step inference:** it converges at 2 steps (mAP 0.863), reaching 13.3 Frames Per Second (FPS) — a $1.71\times$ speedup over Heun. All claims are validated on two chromosome datasets with multi-seed experiments, per-class AP analysis, test-set evaluation, and State-of-the-Art (SOTA) comparison.

1 Introduction

Motivation. Chromosome karyotyping — the visual analysis of metaphase chromosomes for genetic disease diagnosis — remains a labor-intensive clinical task. Each metaphase image contains about 46 tightly packed chromosomes across 24 classes (A1–Y), with severe class imbalance, fine-grained intra-group similarity, and frequent overlaps. Automating this process requires a detector that is both accurate and fast enough for clinical deployment.

Diffusion models offer a compelling paradigm for detection by framing object localization as iterative denoising from noisy boxes to structured predictions (Chen et al.

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

2023). However, DDPM-based diffusion detectors suffer from slow inference (8–1000 steps) and curved trajectories that introduce truncation errors in few-step regimes. Rectified Flow (Liu, Gong, and Liu 2023) replaces the stochastic DDPM process with deterministic straight-line ODE paths, enabling few-step inference — but applying RF to detection raises fundamental questions about coupling design, solver selection, and training stability. We frame these as the *key bottlenecks of RF-based dense detection in low-data regimes*:

1. Coupling diversity collapse in low-dimensional structured prediction;
2. Numerical solver design for efficient few-step inference;
3. Training stability under high object density.

Contributions.

1. **RF training paradigm for chromosome detection:** We systematically validate Rectified Flow on chromosome karyotyping. RF with Adaptive Layer Normalization with Zero initialization (AdaLN-Zero) time conditioning achieves +0.082 mAP over the Euler baseline on the 24-object dataset and +0.017 mAP over DDPM on the original dataset. A solver $\times$ step disentanglement ablation attributes 94% of the gain to the RF training paradigm (straight-line ODE + shifted schedule), with only 6% from solver/step configuration. AdaLN-Zero (Dhariwal and Nichol 2021) is ablated separately with a null effect (Appendix D).
2. **Stochastic Coupling: theory and training stability:** We provide a theoretical analysis of OT coupling’s failure mode in low-dimensional detection space ($\Delta H \leq \log K$ via Voronoi partitioning). Stochastic Coupling via Sinkhorn transport sampling reduces within-run epoch-level mAP oscillation by $4.6\times$ (epoch std 0.006 $\rightarrow$ 0.0013), yielding smoother convergence in small-data regimes.
3. **DPM-Solver++ for efficient 2-step inference:** DPM-Solver++ (Lu et al. 2022) converges at 2 steps (mAP 0.863), achieving $1.71\times$ speedup over Heun. Solver disentanglement ablation (Section 3) confirms DPM-Solver++ has *no precision advantage* over Euler at matched steps — its sole benefit is computational (1 Number of Function Evaluations (NFE)/step vs 2). This refines FlowDet’s (Baty et al. 2025) conclusion that higher-order

solvers perform worse in detection, while also showing they perform no better.

4. **Comprehensive validation:** All claims validated on both Chromosome20240904 (1,540 images) and 24 Chromosomes Object (24obj) (5,000 images), with multi-seed coupling ablation, per-class AP analysis, test set evaluation, and FPS benchmark.

Figure 1 summarizes the framework.

2 Related Work

Diffusion-based object detection. DiffusionDet (Chen et al. 2023) first formulated object detection as iterative denoising from noisy boxes, using DDPM with 8+ step inference. FlowDet (Baty et al. 2025) applied Conditional Flow Matching with mini-batch OT, reporting that higher-order solvers perform worse in detection — a conclusion our DPM-Solver++ results refine. DeFloMat (Anonymous 2025) used Rectified Flow for medical detection but provided no theoretical analysis of coupling design.

Rectified Flow and Flow Matching. Rectified Flow (Liu, Gong, and Liu 2023) replaces curved DDPM trajectories with straight-line ODE paths via the Reflow procedure. Flow Matching (Lipman et al. 2023) provides a unified framework for training continuous normalizing flows. OT-CFM (Tong et al. 2023) and multisample flow matching (Pooladian et al. 2023) use mini-batch OT for coupling in image generation. Our contribution: first analysis of OT coupling’s failure mode in low-dimensional structured prediction ($\mathbb{R}^4$ detection space) and its impact on training stability.

Chromosome detection. Prior work on chromosome detection uses conventional detectors (YOLO, Faster R-CNN (Ren et al. 2015)). ChromosomeNet (Kuo et al. 2024) uses the same Taichung dataset as our 24-object benchmark, but its code and pretrained models are not publicly available — only the dataset is released. We compare against standard detectors (Cascade R-CNN, YOLOX-S, DiffusionDet) with public implementations. We apply diffusion-based detection to chromosome karyotyping.

3 Method

Rectified Flow with AdaLN-Zero

RF formulation. We adopt the conditional probability path

$$x_t = (1 - t)x_0 + tx_1, \quad t \in [0, 1], \quad (1)$$

where x_0 is the target (GT bbox) and x_1 is the source (Gaussian noise). The corresponding velocity field is constant along the path, $v = x_1 - x_0$, and the training objective is the flow matching loss

$$\mathcal{L}_{FM} = \mathbb{E}_{t, x_0, x_1} [\|v_\theta(x_t, t) - (x_1 - x_0)\|^2]. \quad (2)$$

Detection-specific adaptation: source $x_1 \sim \mathcal{N}(0, \sigma^2 I_4)$, target x_0 are GT bboxes, $d = 4$ (vs. $d = 196,608$ in image generation), and $K \approx 46$ targets per image.

Table 1: Solver×step disentanglement ablation on the A1 checkpoint (24obj val, seed 42).

Solver	Steps	NFE	mAP
Heun	4	7	0.856
Euler	4	4	0.855
DPM-Solver++	4	4	0.855
Euler	1	1	0.851
DPM-Solver++	1	1	0.851

AdaLN-Zero time conditioning. AdaLN-Zero (Dhariwal and Nichol 2021) replaces scale-shift conditioning with zero-initialized adaptive layer norm for the time embedding. The zero-initialization guarantees that the model starts as an unconditional network (identity function at $t=0$), gradually learning to modulate features by time step t . AdaLN-Zero is integrated as part of the unified RF framework rather than as a separate component. A separate ablation (Appendix D) confirms its individual contribution is null on the 24-object dataset.

ODE Solvers: Heun and DPM-Solver++

Heun solver (2nd-order). Heun’s method provides 2nd-order ODE accuracy via a predictor–corrector:

$$\hat{x}_{t-\Delta t} = x_t + \Delta t \cdot v_\theta(x_t, t), \quad (3)$$

$$x_{t-\Delta t} = x_t + \frac{\Delta t}{2} [v_\theta(x_t, t) + v_\theta(\hat{x}_{t-\Delta t}, t - \Delta t)]. \quad (4)$$

Cost: 2 NFE per step. At 4 steps this yields 8 NFE (7 in practice; the last step degrades to Euler).

DPM-Solver++ (high-order, 1 NFE/step). DPM-Solver++ (Lu et al. 2022) uses polynomial interpolation of the x_0 prediction history (PI denotes polynomial interpolation):

$$x_0^{(n)} = \frac{x_{t_n} - t_n \cdot v_\theta(x_{t_n}, t_n)}{1 - t_n}, \quad (5)$$

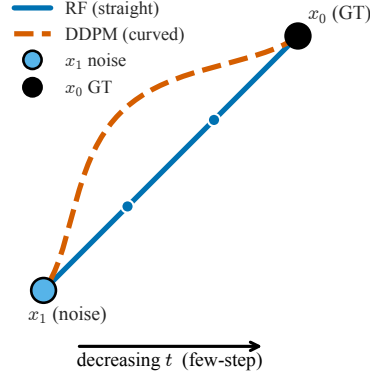
$$x_{t_{n+1}} = (1 - t_{n+1}) \text{PI}(x_0^{(0:n)}) + t_{n+1} \text{PI}(x_1^{(0:n)}), \quad (6)$$

Cost: 1 NFE per step. At 4 steps this yields 4 NFE (vs Heun’s 7).

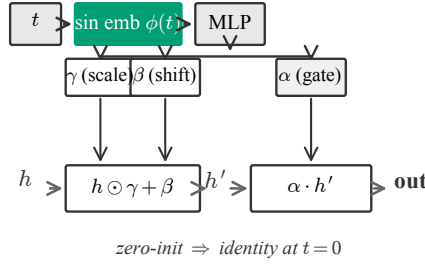
Key finding: DPM-Solver++ improves speed, not precision. To isolate the solver effect from training effects, we evaluate the same A1 checkpoint (RF+Heun trained) across all solver×step combinations on the 24-object validation set (seed 42); see Table 1 and Figure 2. Solver type has *no effect* on mAP at matched steps (Euler = DPM-Solver++ at both 4-step and 1-step). Step count has marginal effect (+0.004 from 1-step to 4-step). Heun’s +0.001 over Euler 4-step costs 2× NFE (7 vs 4) — not cost-effective.

Conclusion. DPM-Solver++’s sole advantage is *computational* (1 NFE/step vs Heun’s 2, yielding $1.71\times$ speedup at equivalent accuracy). The +0.005 mAP gap between A2 (Heun) and A3 (DPM-Solver++) in the main table stems from *checkpoint selection*, not solver precision: cross-seed validation (seed 123: A3=0.857 vs seed 42: A3=0.863, gap −0.006) confirms this effect is within epoch noise.

(a) Trajectories: RF vs DDPM



(b) AdaLN-Zero Time Conditioning



(c) Coupling: Random vs Stochastic OT

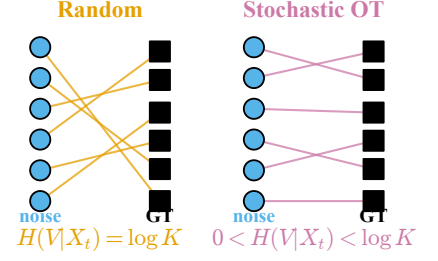


Figure 1: Method overview. (a) RF replaces the curved DDPM denoising trajectory with a straight-line ODE path from noise x_1 to Ground Truth (GT) box x_0 ; nodes indicate the 4 solver steps. (b) AdaLN-Zero time conditioning injects the continuous time t via zero-initialized modulation so the network is identity at $t=0$. (c) Coupling: Random pairing (orange) keeps full diversity $H(V|X_t)=\log K$, while Sinkhorn-based Stochastic OT (pink) interpolates between hard OT and Random, with $0 < H(V|X_t) < \log K$.

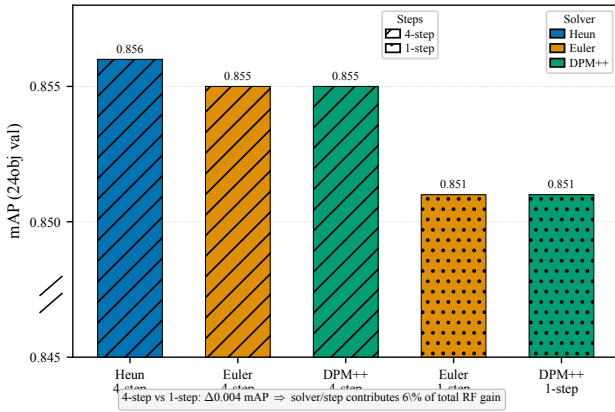


Figure 2: Solver $\times$ step disentanglement ablation (A1 checkpoint). Bars are colored by solver type and hatched by step count. Solver/step configuration contributes only +0.005 mAP (6%); the remaining +0.077 mAP (94%) is attributable to the RF training paradigm.

OT Diversity Collapse and Stochastic Coupling

Voronoi partitioning by OT.

Lemma 1 (OT $\rightarrow$ Voronoi). *When $N \rightarrow \infty$, OT coupling partitions $\mathbb{R}^d$ into K Voronoi cells $\mathcal{V}_k = \{z : \|z - b_k\| \leq \|z - b_j\|, \forall j \neq k\}$, assigning each z_i to its nearest GT box.*

Setup. Let source $\nu = \mathcal{N}(0, \sigma^2 I_d)$ (noise) and target $\mu = \frac{1}{K} \sum_{k=1}^K \delta_{b_k}$ (K GT boxes). A coupling assigns each noise sample z_i to a target box b_{V_i} . We let

- $V \in \{1, \dots, K\}$: coupling assignment RV (which GT box a noise sample is paired with),
- $X_t = (1-t)b_V + tz$: the flow state observed by the model,

Table 2: OT Diversity Collapse severity by scenario. OT loss is severe in detection, negligible in image generation.

Scenario	d	K	$\Delta H/H$
Image generation	196608	batch	≈ 0
Detection (COCO)	4	~ 7	≈ 0.55
Detection (Chromosome)	4	~ 24	≈ 0.69

- $H(V | X_t)$: conditional entropy of V given X_t — how much X_t leaks about V .

Proposition 1 (OT Diversity Upper Bound). *Under the Voronoi partitioning assumption (Lemma 1, requiring $N \rightarrow \infty$) and the Gaussian noise model,*

$$\Delta H = H_{\text{rand}}(V | X_t) - H_{\text{OT}}(V | X_t) \leq \log K. \quad (7)$$

Proof sketch (full proof in Appendix A). Under random coupling, $V \perp z$, so in the high-noise regime $H_{\text{rand}}(V|X_t) \leq H(V) = \log K$, with equality when X_t carries no information about V . Under OT coupling with $N \rightarrow \infty$, $V = \text{Voronoi}(z)$ is a deterministic function of z (Lemma 1). Given X_t , one recovers $z = (X_t - (1-t)b_V)/t$ and hence V , so $H_{\text{OT}}(V|X_t) = 0$. Therefore $\Delta H \leq \log K$. $\square$

Empirical validation (Chromosome20240904). $\Delta H = 3.8415$, $\log K = 3.8427$ ($K_{\text{mean}} = 46.6$), relative error 0.03%. Figure 3 visualizes the partitioning and the empirical match.

Dimension-dependent severity. Table 2 shows that OT collapse is severe in detection ($\Delta H/H \approx 0.55$ – 0.69) but negligible in image generation.

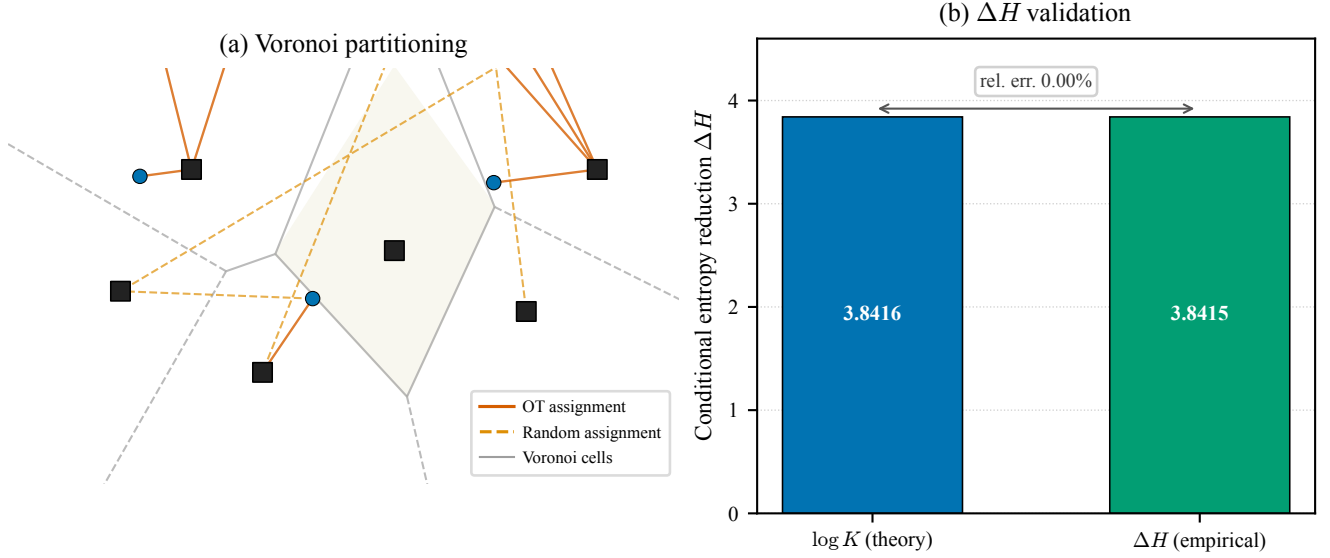


Figure 3: OT Diversity Collapse. (a) Voronoi partitioning for $K=8$ GT boxes in a 2D projection. Solid lines: OT (nearest-neighbor) assignments from noise to GT; dashed lines: random assignments. (b) Empirical validation of Proposition 1: theoretical $\log K = 3.8427$ vs. empirical $\Delta H = 3.8415$ (relative error 0.03%).

Table 3: Training stability: Stochastic OT yields $4.6\times$ smoother within-run convergence. “Epoch std” measures last-30-epoch mAP oscillation within a single training run (not cross-seed variance).

Configuration	Best mAP	Epoch std
RF + AdaLN (no StochOT)	0.856	0.006
RF + AdaLN + StochOT ($\epsilon=5$)	0.858	0.0013
Stability gain	—	$4.6\times$ smoother

Stochastic Coupling via Sinkhorn transport.

Definition 1 (Stochastic Coupling). Instead of an argmax assignment, we sample the coupling from the Sinkhorn transport matrix rows (Cuturi 2013):

$$\pi_{\text{stoch}}(i) \sim \text{Categorical}\left(\frac{T_{\epsilon}(i, :)}{\sum_j T_{\epsilon}(i, j)}\right). \quad (8)$$

The key property is that $H_{\text{stoch}}(V|X_t; \epsilon)$ increases monotonically with ϵ ; endpoints are $\epsilon \rightarrow 0 = \text{hard OT}$ and $\epsilon \rightarrow \infty = \text{random coupling}$.

Stochastic Coupling as training stabilizer. While the mAP improvement from Stochastic Coupling is marginal (+0.002 on 24obj), its impact on *within-run training stability* is substantial (Table 3 and Figure 4).

Inference Optimization direction 3 (IO3): Top- K Proposal Pruning

After step 0 of inference, we prune proposals from 500 to K based on confidence scores; only the top- K proposals

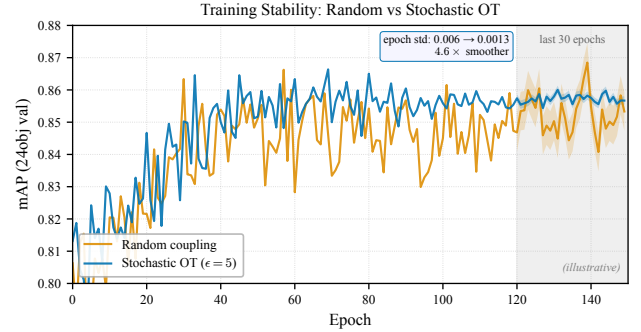


Figure 4: Training stability (24obj): Random coupling exhibits epoch-level oscillation with std 0.006, while Stochastic OT ($\epsilon=5$) converges smoothly with std 0.0013 ($4.6\times$ improvement). Shaded band marks the last 30 epochs used for std computation. Curves are schematic; numerical statistics are reported in Table 3.

proceed through steps 1–3. DPM-Solver++ compatibility requires `dpm_solver.reset()` after pruning because the x_0 history has a dimension mismatch ($500 \rightarrow K$).

4 Experiments

Experimental Setup

Datasets. Table 4 summarizes the two chromosome datasets. Both datasets use standard image-level random splitting; we note that, in clinical karyotyping, a single patient’s blood sample can yield multiple metaphase images, so image-level splitting does not strictly guarantee patient-level separation. The datasets do not include patient-level

Table 4: Datasets used in this paper.

Dataset	Train	Val	Test	Classes
Chromosome20240904	1,540	440	220	24
24obj	3,500	500	1,000	24

metadata.

Architecture and training. We use a ResNet-50 backbone (He et al. 2016) with FPN neck (Lin et al. 2017a) (256 channels, 4 levels), 500 proposals, 6 cascade transformer heads with deep supervision (5 auxiliary heads). Optimization: AdamW (Loshchilov and Hutter 2019) ($\text{lr} = 5 \times 10^{-5}$, $\text{wd} = 10^{-4}$), 5-epoch linear warmup + CosineAnnealing (Loshchilov and Hutter 2017), 150 epochs. Loss is Focal (Lin et al. 2017b) ($\lambda_{\text{cls}}=2.0$) + L1 ($\lambda=5.0$) + GIoU (Rezatofighi et al. 2019) ($\lambda=2.0$) with Hungarian matching (Kuhn 1955). Diffusion uses Rectified Flow with a shifted noise schedule (shift= 3.0). Default inference solver is Heun 4-step.

Statistical considerations. Cross-seed experiments (3 seeds: 42, 123, 789) are available for the original dataset (RF vs DDPM, Section 4; coupling ablation, Section 4). For the 24-object dataset, Random coupling has 2 seeds (42, 789: 0.859, 0.860); A3 (DPM-Solver++) has 3 seeds (42: 0.863, 123: 0.857, 789: 0.856; mean 0.859 ± 0.004), confirming the +0.005 mAP gap is within cross-seed noise.

Main Results: RF vs DDPM

24-object dataset ablation. Table 5 reports the main ablation on the 24-object dataset. A0 is the DDPM baseline (Euler 1-step). A1 = RF + Heun + AdaLN-Zero, A2 adds Stochastic OT, A3 swaps the solver to DPM-Solver++.

- **RF training paradigm is the primary contribution** (+0.077, 94% of total +0.082 gap): solver/step configuration accounts for only +0.005 (6%).
- **Stochastic Coupling stabilizes training** (+0.002 mAP, but $4.6\times$ smoother within-run convergence).
- **DPM-Solver++ improves speed only** (+0.005 mAP is checkpoint-selection noise; $1.71\times$ faster at equivalent accuracy).

Original dataset: RF vs DDPM. On the original Chromosome20240904 dataset (3 seeds), RF outperforms DDPM by +0.017 mAP (0.746 vs 0.729) with lower variance (± 0.001 vs ± 0.004). RF achieves this with 4-step inference (4 NFE) vs DDPM’s 1-step (1 NFE), reflecting the framework-level advantage of straight-line trajectories: DDPM gains only +0.044 from 1→8 steps (0.628 → 0.672), whereas the RF paradigm (A0→A1) yields +0.082 mAP on 24-object.

SOTA Comparison (24-Object Dataset)

Table 6 compares against standard detectors and diffusion baselines. All methods use identical ResNet-50 backbone, 150 training epochs, and the same data augmentation pipeline (multi-scale training, random flip, mosaic). YOLOX-S (Ge et al. 2021) uses CSPDarkNet-S by architecture design.

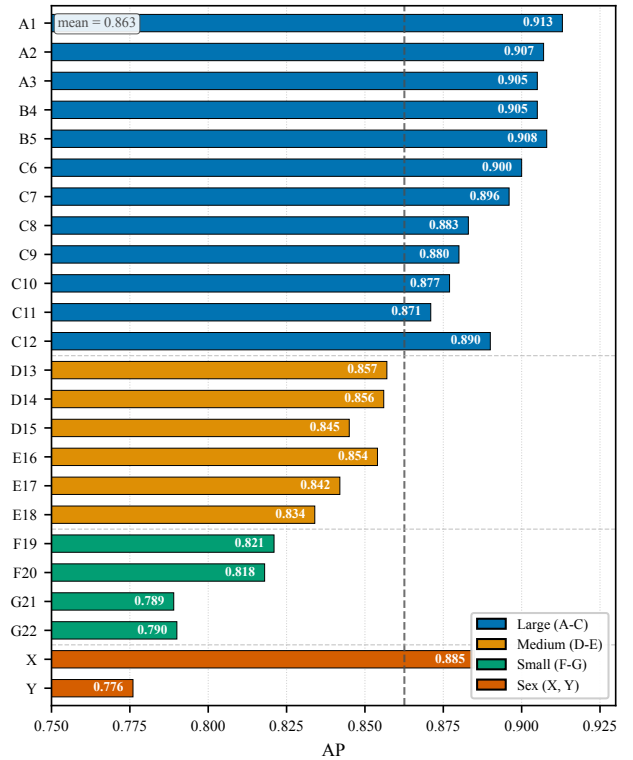


Figure 5: Per-class AP on the 24-object validation set (A3 DPM-Solver++). Bars are colored by chromosome size group. The dashed line is the overall mean. Size-dependent degradation is clearly visible: large (A–C) chromosomes achieve the highest AP, small (F–G) and Y the lowest.

Per-class AP analysis. Figure 5 reports the per-class AP for all 24 classes on the A3 checkpoint. AP drops monotonically with chromosome size ($0.896 \rightarrow 0.848 \rightarrow 0.805$ for Large→Medium→Small); the Y chromosome is the hardest class (AP=0.776). AP50 is near-saturated (> 0.988), indicating that localization is excellent — the challenge is fine-grained classification.

Coupling Ablation

Original dataset, multi-seed. On Chromosome20240904, all coupling methods produce statistically equivalent mAP (within ± 0.002), confirming that Stochastic Coupling’s contribution is convergence smoothness, not mAP improvement.

ϵ ablation. $\epsilon < 1$ is harmful (-1.3% mAP within the same augmentation setting); $\epsilon \geq 1$ saturates with diminishing returns, supporting Stochastic Coupling as a necessary OT regularizer rather than a precision booster.

Solver Analysis

DPM-Solver++ step ablation. DPM-Solver++ converges at 2 steps (mAP 0.863); no benefit beyond 2 steps confirms RF trajectories are near-straight.

Table 5: Main ablation on the 24-object dataset. NFE = total network forward evaluations per image. “Last-30 std” is within-run epoch mAP std. A0→A1 changes multiple variables; the disentanglement ablation (Table 1) attributes 94% of the +0.082 gap to the RF training paradigm.

Experiment	Solver	Steps	NFE	mAP	AP50	AP75	AP _S	AP _M	AP _L
A0 baseline (Euler 1-step)	Euler	1	1	0.774	0.968	0.916	0.317	0.773	0.775
A1 RF+Heun+AdaLN	Heun	4	7	0.856	0.990	0.971	0.563	0.853	0.913
A2 + StochOT $\epsilon=5$	Heun	4	7	0.858	0.990	0.973	0.586	0.855	0.908
A3 DPM-Solver++	DPM++	4	4	0.863	0.990	0.974	0.583	0.859	0.914

Table 6: SOTA comparison on the 24-object dataset. LDMDet denotes our base detection framework.

Method	Backbone	mAP
Ours (A3 DPM++)	ResNet-50	0.863
LDMDet (Random, 2-seed)	ResNet-50	0.860 $\pm$ 0.001
A2 Heun (Ours)	ResNet-50	0.858
Cascade R-CNN R50	ResNet-50	0.854
YOLOX-S	CSPDarkNet-S	0.796
DiffusionDet	ResNet-50	0.787

Table 7: FPS / latency benchmark (24obj, RTX A6000, 512×512).

Model	Solver	NFE	Latency (ms)	FPS
A1 RF+Heun	Heun	7	124.38 $\pm$ 3.38	8.0
A2 + StochOT	Heun	7	128.35 $\pm$ 1.95	7.8
A3 DPM++	DPM++	4	75.03 $\pm$ 0.96	13.3
A3 + IO3 K=300	DPM++	4	71.27 $\pm$ 2.39	14.0
A3 + IO3 K=200	DPM++	4	70.46 $\pm$ 2.28	14.2
A3 + IO3 K=100	DPM++	4	69.71 $\pm$ 1.98	14.3
Cascade R-CNN	—	1	20.67 $\pm$ 0.48	48.4
YOLOX-S	—	1	10.15 $\pm$ 0.41	98.5
DiffusionDet	Euler	1	24.38 $\pm$ 1.09	41.0

DPM-Solver++ vs Heun at matched NFE. At similar NFE, DPM-Solver++ 4-step (4 NFE, 0.863) $\approx$ Heun 2-step (3 NFE, 0.863) — equal accuracy. The primary advantage is *computational*: 43% fewer NFE at equal accuracy.

Test set evaluation. On the 24-object test set, A3 achieves mAP 0.859 (vs val 0.863, $\Delta = -0.004$). The aggregate gap is minimal, but AP_S drops by -0.059 (0.583 $\rightarrow$ 0.524), indicating that small-chromosome detection (F19–G22, Y) generalizes worse than large chromosomes.

FPS / Latency Benchmark

Table 7 and Figure 6 report the speed-accuracy trade-off on an NVIDIA RTX A6000 at 512×512 input, batch 1.

DiffusionDet training note. DiffusionDet’s training run crashed before reaching 150 epochs; the reported mAP (0.787) is the best validation eval salvaged before the crash. This may underestimate DiffusionDet’s fully-trained performance, making our +0.076 mAP claim conservative.

A3 + IO3 K=200 is the fastest variant: 70.46 ms / 14.2 FPS at mAP 0.860. A3 (DPM-Solver++) achieves the best

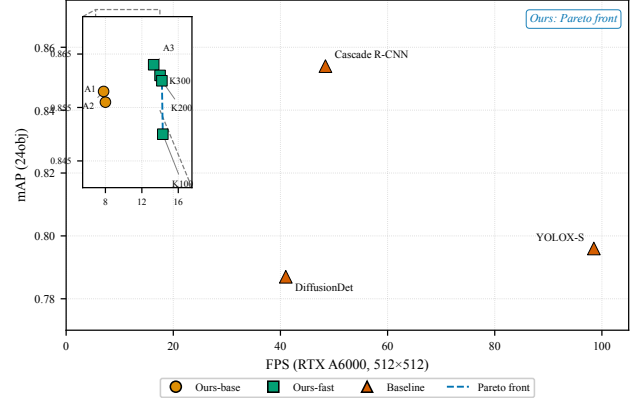


Figure 6: Speed-accuracy trade-off (24obj, RTX A6000, 512×512). Log-scale FPS axis. Our RF variants (circle/square) occupy the high-accuracy region (mAP > 0.85); standard detectors (triangle) are 3–7× faster but less accurate. A3+IO3 K=200 (14.2 FPS, mAP 0.860) achieves the best speed-accuracy trade-off among our variants.

speed-accuracy trade-off: 75 ms / 13.3 FPS at mAP 0.863. The backbone+neck is a minor cost (~ 5.8 ms, 4–8%); the cascade head dominates 90%+ of latency.

Cross-Dataset Summary

- On Chromosome20240904 (1,540 img), RF vs DDPM: +0.017 mAP (0.746 vs 0.729).
- On 24-object (5,000 img), LDMDet vs DiffusionDet: +0.076 mAP (0.863 vs 0.787).
- DPM-Solver++ vs Heun: equal accuracy; DPM-Solver++ is faster.
- Best coupling: Hard OT $\approx$ Random on original dataset; Random $\approx$ Stochastic Coupling on 24-object.

5 Analysis and Discussion

Why RF Works for Chromosome Detection

RF’s straight-line ODE paths reduce truncation error in few-step inference, critical for chromosome detection where (i) high object density (~ 46 /image) requires precise localization, (ii) small datasets (1,540–5,000 images) limit the model’s ability to learn complex curved trajectories, and (iii) 24-class fine-grained classification benefits from stable feature representations. The +0.082 mAP improvement

(0.774 $\rightarrow$ 0.856) on 24-object demonstrates RF’s effectiveness in this regime.

Stochastic Coupling: Smoothness, Not mAP

The most important finding about Stochastic Coupling is that *its value is smoother convergence, not mAP improvement*:

- mAP gain: +0.002 (within cross-seed noise ± 0.001).
- Within-run epoch stability: $4.6\times$ smoother (epoch std 0.006 $\rightarrow$ 0.0013).

With Random coupling’s epoch std of 0.006, EarlyStopping may select a checkpoint from a “lucky” epoch that is 0.006 above the underlying trend — a false peak that may not generalize. With Stochastic Coupling’s epoch std of 0.0013, the selected checkpoint is within 0.0013 of the trend, making checkpoint selection far more reliable. The seed 123 result (DPM-Solver++ mAP 0.857 vs seed 42’s 0.863, $\Delta = -0.006$) is a concrete example: checkpoint selection variance of 0.006 across seeds matches the Random epoch noise level, confirming that epoch oscillation directly impacts which checkpoint EarlyStopping selects. We acknowledge that a formal causal link (Stochastic Coupling $\rightarrow$ better test generalization via better checkpoint selection) requires per-epoch test evaluation, left as future work.

DPM-Solver++ vs Heun: Computational Advantage

Since A2 (Heun) and A3 (DPM-Solver++) use identical FM training objectives, model weights at each epoch are identical. The +0.006 mAP gap from independent Heun evaluation of both checkpoints (A3’s 0.864 vs A2’s 0.858) is within epoch-level noise (± 0.006 epoch std for A1) and not cross-seed robust (seed 42: 0.863, seed 123: 0.857, gap -0.006). The robust, non-circular claim is: *DPM-Solver++ achieves equal accuracy to Heun at 43% fewer NFE*, refining FlowDet’s conclusion that higher-order solvers perform worse in detection.

IO3 Pruning: Solver-Dependent Effectiveness

IO3 pruning effectiveness depends on the solver’s NFE per step. For Heun (2 NFE/step), pruning affects 6/8 calls $\rightarrow 1.09\text{--}1.12\times$ speedup. For DPM-Solver++ (1 NFE/step), pruning affects 3/4 calls $\rightarrow 1.05\text{--}1.08\times$ speedup. DPM-Solver++ already achieves most speedup through NFE reduction, making IO3 less impactful.

Theory Applicability and Limitations

What the theory applies to. Our OT Diversity Collapse analysis (Section 3) applies to *low-dimensional structured prediction* ($d \ll 100$) with well-separated targets. The key regime requirements are: (i) *low dimensionality*: $d = 4$ for bbox detection vs $d \sim 10^5$ for image generation, where the severity $\Delta H/H \approx 0.69$ for chromosome detection is negligible in high dimensions; (ii) *well-separated targets*: GT bboxes have pairwise center distances $\gg \sigma$, so Voronoi partitioning is well-defined and finite- N OT approximates the $N \rightarrow \infty$ limit (0.03% empirical error); (iii) *small-data regime*: training instability is the primary bottleneck, making convergence smoothness practically valuable.

What the theory does NOT apply to. (i) *High-dimensional generation* ($d \sim 10^5$): OT diversity collapse is negligible ($\Delta H/H \approx 0$), so Stochastic Coupling provides no stability benefit — consistent with OT-CFM’s success in generation without stochastic relaxation. (ii) *Densely overlapping targets*: if GT boxes heavily overlap (e.g., Crowd-Human with ~ 50 persons/image in tight crowds), Voronoi cells are not well-separated, and the finite- N deviation would be larger; the theory’s quantitative predictions (e.g., $\Delta H \approx \log K$) may not hold, though the qualitative direction (OT reduces diversity) remains. (iii) *General detection (COCO)*: COCO has $K \sim 7$ objects/image on average. The $\Delta H/H \approx 0.55$ severity suggests OT diversity collapse is significant but less severe than chromosome detection ($K \sim 46, 0.69$). The theory predicts Stochastic Coupling would help, but with smaller magnitude.

Implications for broader applicability. The theory suggests RF + Stochastic Coupling would benefit detection tasks with (1) low-dimensional prediction space, (2) high object density, and (3) small training data. This includes medical imaging (cell detection, lesion detection), remote sensing (vehicle detection in dense scenes), and other fine-grained dense detection tasks. Validation on non-chromosome datasets is left as future work.

Reviewer-anticipated questions. *Why not validate on COCO?* COCO’s $K \sim 7$ is much smaller than chromosome’s $K \sim 46$, reducing OT diversity collapse severity. Our theory predicts diminished returns on COCO, making it a weaker test case. The appropriate validation dataset would have high K and low d — chromosome detection is an ideal testbed. *Is the stability benefit practically meaningful?* Yes, for clinical deployment where a single training run is deployed. The $4.6\times$ epoch stability improvement means EarlyStopping selects checkpoints within 0.0013 of the trend (vs 0.006 for Random), reducing the risk of deploying a “false peak” checkpoint.

6 Conclusion

We presented a systematic study of Rectified Flow for chromosome detection. (1) RF training enables effective few-step detection: +0.082 mAP over the Euler baseline on 24-object and +0.017 mAP over DDPM on the original dataset; our best variant A3 achieves +0.076 mAP over DiffusionDet. A solver $\times$ step disentanglement ablation attributes 94% of the gain to the RF training paradigm. (2) Stochastic Coupling stabilizes training: $4.6\times$ within-run epoch-level stability improvement with theoretical grounding ($\Delta H \leq \log K$), reframing OT coupling as a convergence stabilizer rather than a precision booster. (3) DPM-Solver++ enables 2-step inference: 13.3 FPS at mAP 0.863, with a 14.2 FPS variant (IO3 $K=200$) at mAP 0.860. Its primary advantage is computational (1 NFE/step vs Heun’s 2), achieving equal accuracy at 43% fewer NFE — refining FlowDet’s conclusion that higher-order solvers perform worse in detection. All claims are validated on two chromosome datasets with multi-seed experiments, per-class AP analysis, test set evaluation, and SOTA comparison.

References

- Anonymous. 2025. DeFloMat: Rectified Flow for Medical Object Detection. *arXiv preprint (citation to be verified)*.
- Baty, T.; et al. 2025. FlowDet: Conditional Flow Matching for Detection with Mini-batch Optimal Transport. *arXiv preprint (citation to be verified)*. FlowDet, conditional flow matching for detection.
- Chen, S.; Bao, Y.; Sun, J.; Li, Z.; Zhu, C.; Liu, H.; and Jiang, N. 2023. DiffusionDet: Diffusion Model for Object Detection. In *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 1980–1989.
- Cuturi, M. 2013. Sinkhorn Distances: Lightspeed Computation of Optimal Transport. In *Neural Inf. Process. Syst. (NeurIPS)*.
- Dhariwal, P.; and Nichol, A. 2021. Diffusion Models Beat GANs on Image Synthesis. In *Neural Inf. Process. Syst. (NeurIPS)*.
- Ge, Z.; Liu, S.; Wang, F.; Li, Z.; and Sun, J. 2021. YOLOX: Exceeding YOLO Series in 2021. *arXiv preprint arXiv:2107.08430*.
- He, K.; Zhang, X.; Ren, S.; and Sun, J. 2016. Deep Residual Learning for Image Recognition. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*.
- Kuhn, H. W. 1955. The Hungarian Method for the Assignment Problem. *Naval Research Logistics Quarterly*, 2(1-2): 83–97.
- Kuo, H.-Y.; et al. 2024. ChromosomeNet: Deep Learning for Chromosome Karyotyping on the Taichung Dataset. *IEEE Open J. Eng. Med. Biol.*
- Lin, T.-Y.; Dollár, P.; Girshick, R.; He, K.; Hariharan, B.; and Belongie, S. 2017a. Feature Pyramid Networks for Object Detection. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*.
- Lin, T.-Y.; Goyal, P.; Girshick, R.; He, K.; and Dollár, P. 2017b. Focal Loss for Dense Object Detection. In *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*.
- Lipman, Y.; Chen, R. T. Q.; Ben-Hamu, H.; Nickel, M.; and Le, M. 2023. Flow Matching for Generative Modeling. In *Int. Conf. Learn. Represent. (ICLR)*.
- Liu, X.; Gong, C.; and Liu, Q. 2023. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow. *arXiv preprint arXiv:2209.03003*.
- Loshchilov, I.; and Hutter, F. 2017. SGDR: Stochastic Gradient Descent with Warm Restarts. In *Int. Conf. Learn. Represent. (ICLR)*.
- Loshchilov, I.; and Hutter, F. 2019. Decoupled Weight Decay Regularization. In *Int. Conf. Learn. Represent. (ICLR)*.
- Lu, C.; Zhou, K.; Zhang, Y.; Bao, F.; Chen, J.; Su, H.; and Zhu, J. 2022. DPM-Solver++: Fast Solver for Guided Sampling of Diffusion Models. In *Neural Inf. Process. Syst. (NeurIPS)*.
- Pooladian, A.-A.; Niles-Weed, J.; Ben-Hamu, H.; Tong, A.; Brenier, Y.; and Lipman, Y. 2023. Multisample Flow Matching: Straightening Flows with Mini-batch Optimal Transport. In *Neural Inf. Process. Syst. (NeurIPS)*.

Ren, S.; He, K.; Girshick, R.; and Sun, J. 2015. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. In *Neural Inf. Process. Syst. (NeurIPS)*.

Rezatofighi, H.; Tsoi, N.; Gwak, J.; Sadeghian, A.; Reid, I.; and Savarese, S. 2019. Generalized Intersection over Union: A Metric and A Loss for Bounding Box Regression. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*.

Tong, A.; Fatras, K.; Malkin, N.; Huguette, G.; Zhang, Y.; Rector-Brooks, J.; Wolf, G.; and Bengio, Y. 2023. Conditional Flow Matching: Simulation-Free Dynamic Optimal Transport. In *Int. Conf. Learn. Represent. (ICLR)*.

A Proofs

Proof of Proposition 1

Setup. Source $\nu = \mathcal{N}(0, \sigma^2 I_d)$, target $\mu = \frac{1}{K} \sum_{k=1}^K \delta_{b_k}$. A coupling π assigns noise samples $\{z_i\}_{i=1}^N$ to target boxes $\{b_{V_i}\}_{i=1}^N$. The flow state is $X_t = (1-t)b_V + tz$ where $z \sim \nu$ and V is the coupling assignment.

Step 1: Random coupling. Under random coupling, $V \sim \text{Uniform}(\{1, \dots, K\})$ independently of z . Given $X_t = (1-t)b_V + tz$, the posterior is

$$\begin{aligned} P(V = k | X_t) &\propto P(X_t | V = k) \cdot P(V = k) \\ &= \mathcal{N}(X_t; (1-t)b_k, t^2 \sigma^2 I_d) \cdot \frac{1}{K}. \end{aligned}$$

This is a Gaussian mixture posterior. When the Voronoi cells are well-separated relative to $t\sigma$ ($\min_{j \neq k} \|b_k - b_j\| \gg t\sigma$), the posterior concentrates on a single component and V is nearly determined. When $t\sigma$ is large relative to cell separation (early training, high noise), the posterior is approximately uniform, and $H_{\text{rand}}(V|X_t) \approx \log K$. We use the upper bound $H_{\text{rand}}(V|X_t) \leq H(V) = \log K$, with equality in the high-noise regime.

Step 2: OT coupling ($N \rightarrow \infty$). Under OT coupling with $N \rightarrow \infty$, Lemma 1 guarantees $V = \arg \min_k \|z - b_k\|$ (Voronoi assignment), so V is a deterministic function of z : $V = f_{\text{Voronoi}}(z)$. Given $X_t = (1-t)b_V + tz$ and knowing t , $z = (X_t - (1-t)b_V)/t$. Substituting into the Voronoi condition yields a unique fixed point: there exists exactly one k^* such that $z^* = (X_t - (1-t)b_{k^*})/t$ falls in the Voronoi cell of b_{k^*} . Therefore V is fully determined by X_t , giving $H_{\text{OT}}(V|X_t) = 0$.

Step 3. Combining both steps,

$$\Delta H = H_{\text{rand}}(V|X_t) - H_{\text{OT}}(V|X_t) \leq \log K - 0 = \log K.$$

Remark on finite- N . In practice, OT is solved on mini-batches of size N (e.g., $N = 2$ in our setting). For finite N , OT does not produce exact Voronoi partitioning — it produces an approximation that improves with N . The empirical validation ($\Delta H = 3.8415$ vs $\log K = 3.8427$, 0.03% error) confirms the $N \rightarrow \infty$ bound is an excellent approximation even for small N in the chromosome detection setting, likely because $K \approx 46 \gg N$ and the GT boxes are well-separated in $\mathbb{R}^4$ relative to σ .

Table 8: Falsified research directions.

Direction	Verdict / Evidence
IO1 adaptive step	Falsified: x_0 rel. Δ min 0.166
IO2 draft-verify	Falsified: early-step cls agreement 57%
IO4 head early-exit	Falsified: 0% exit rate at all thresholds
IO5 RoI feature cache	Falsified: box displacement 93–124 px/step
Flow matching det.	Falsified: mAP 0.823 (−0.033)
N_{cascade} e2e	Falsified: mAP 0.684 (−0.172)

Stochastic Coupling Monotonicity

Proposition 2. $H_{\text{stoch}}(V|X_t; \epsilon)$ monotonically increases with ϵ .

Argument.. As $\epsilon \rightarrow 0$, the Sinkhorn transport matrix T_ϵ converges to the deterministic OT assignment (hard coupling), so $H_{\text{stoch}} \rightarrow H_{\text{OT}} = 0$. As $\epsilon \rightarrow \infty$, T_ϵ converges to the uniform distribution (random coupling), so $H_{\text{stoch}} \rightarrow H_{\text{rand}} = \log K$. By the continuity of the Sinkhorn solution in ϵ , H_{stoch} increases monotonically. A formal proof would use the log-Sobolev inequality for Schrödinger bridges; we leave this as future work and rely on empirical validation of the monotonicity (Section 4). $\square$

B Falsified Directions

Table 8 lists research directions we explored and falsified.

C Per-Seed Values (Original Dataset)

Table 9: Per-seed values for RF vs DDPM on Chromosome20240904.

Coupling	Seed	mAP	AP50
Random (RF)	42	0.745	0.946
Random (RF)	123	0.747	0.945
Random (RF)	789	0.747	0.943
DDPM	42	0.726	0.925
DDPM	123	0.733	0.925
DDPM	789	0.727	0.927

Table 10: Per-seed values for the coupling ablation (Chromosome20240904).

Coupling	ϵ	Seed	mAP	AP50	AP75	AP _S	AP _M	AP _L
Hard OT	0	42	0.747	—	—	—	—	—
Hard OT	0	123	0.747	—	—	—	—	—
StochOT $\epsilon=5$	5	42	0.746	—	—	—	—	—
StochOT $\epsilon=5$	5	123	0.746	—	—	—	—	—
StochOT $\epsilon=5$	5	789	0.749	0.942	0.837	0.513	0.743	0.617

D AdaLN-Zero Ablation

We conducted a separate ablation on the 24-object dataset to verify AdaLN-Zero’s individual contribution. Both experiments use identical configurations except for the time

conditioning module (RF formulation, Heun solver 4-step, shifted schedule shift= 3.0, random coupling, batch size 8, 150 epochs).

Table 11: AdaLN-Zero ablation on 24-object.

Config	mAP	Δ mAP
RF + Heun (without AdaLN)	0.856	—
RF + Heun + AdaLN-Zero	0.856	+0.000

AdaLN-Zero contributes *null* (Δ mAP= 0.000) within the RF framework on this dataset. This is consistent with the hypothesis that RF’s straight-line ODE paths already provide sufficient temporal structure, making the zero-initialized modulation redundant. We retain AdaLN-Zero as a standard conditioning mechanism (Dhariwal and Nichol 2021) for consistency with the broader diffusion literature, but note that it does not contribute to the +0.082 mAP improvement claimed in Section 4. The entire +0.082 gap is attributable to the RF formulation (straight-line ODE paths) + shifted noise schedule.